

The University of Texas at Austin Invitational

October 30th, 2021



Food Science – Division B

Exam Booklet

KEY

Written by Jaehyun Ahn [JA] and Laura Tan [LT]

(Jaehyun.Ahn.0730@Outlook.com & Laura.p.Tan@gmail.com)

Instructions:

- **Do not begin the test until instructed to do so.**
- For this event, you are allowed:
- You will have 50 minutes to complete the exam.
- For any calculation questions, please show your work! Partial credits will be awarded when necessary.
- This test consists of four sections: (1) Multiple Choice Questions (2) Free Response Questions (3) True and False (4) Hands-On Lab Practical.
- For Section 1, there may be multiple choice questions that have more than one correct answer(s). These will be indicated with SELECT ALL THAT APPLY. (Partial Credits will be awarded)
- Multiple choice questions with more than one correct answers are worth 2 points while those with one correct answers are worth 1 point.
- For Section 4, please **carefully** read the directions stated on the exam.
- Tie breakers will be labeled as [TB #]

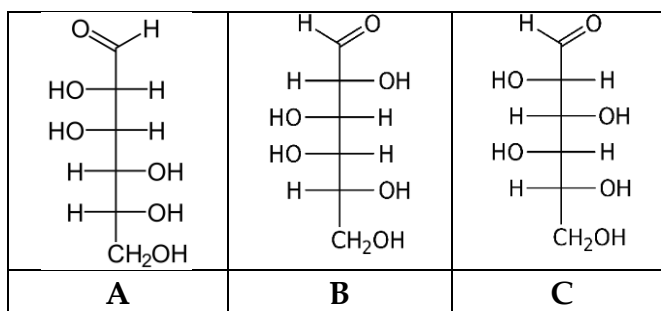
Team Name: _____

Team # _____

Team Members: _____

Section I: Multiple Choice Questions

Use the following information for Question 1 and 2.



The diagram above shows **three isomers** of the sugar molecule **glucose**. Although these molecules have the same chemical formula and composition as glucose, they have different bonding patterns.

- 1.) Given the information above, determine which of the following is (are) **correct** (Select all that apply).

[JA]

- A. All three molecules have *C:H:O* ratio of 1:2:1.
- B. They are all reducing sugars.
- C. They are all disaccharides.
- D. They are all ketoses.
- E. They are all hexoses.

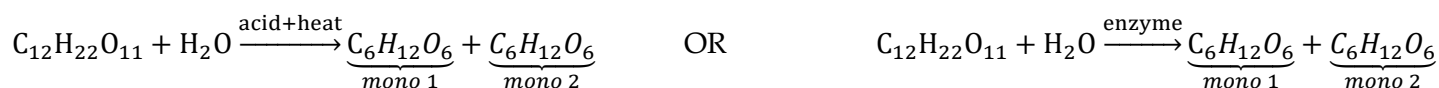
- 2.) Which of the following sugars above is **galactose**? [JA]

- A. A **B. B** C. C D. A & B E. B & C

Use the following information for Questions 3 and 4.

A type of liquid sugar that you may have consumed before is called *invert sugar*. Invert sugar is created when a solution of *sucrose* is treated by an acid or an enzyme. There are two ways to complete the formation of invert sugar: (1) *acid hydrolysis* and (2) *enzyme hydrolysis*. Each method has different requirements and procedures.

The reaction can be summarized by the chemical reaction below



*For the purpose of this exam, *monosaccharide* has been abbreviated to *mono**

- 3.) Which of the following is (are) true about the reaction above? (Select all the apply) [JA] [TB #1]

- A) When the products are tested with Benedicts solutions, the color will turn brick-red.
- B) If too much acid is added, the product will be soft and runny.
- C) A slow rate of heating will increase the chance of inversion.
- D) The products of the reactions are not isomers of each other
- E) The addition of lemon juice when heating sucrose can accelerate the chemical reaction.

- 4.) As shown above, one of the reactions uses enzymes to catalyze the reaction. Which of the following is (are) **true** regarding the enzyme in the reaction? (Select all that apply) [JA]
- A) It can be categorized as a lyase.
 - B) It is consumed and used up during the reaction.
 - C) It accelerates the chemical reaction process
 - D) The role of the enzyme is to break the glycosidic linkage between Carbon 1 of monosaccharide 1 and Carbon 2 of monosaccharide 2.
 - E) Its main component is a protein.

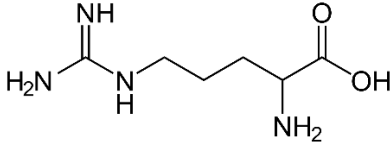
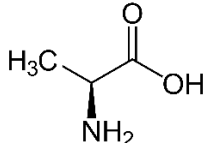
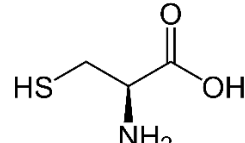
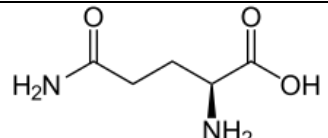
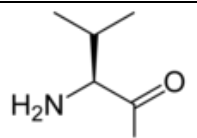
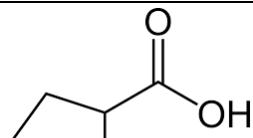
Use the following information for Questions 5 to 7.

Once a food has been made, it must undergo a preservation process to prevent pathogens from forming. Without the food preservation process, the food may become the target of bacterial spoilage and lose its nutritive/textural characteristics. Thus, companies must employ food preservation processes to prevent or delay the spoilage of food.

- 5.) Which of the following is (are) a type of food preservation techniques? (Select all that apply) [JA]
- A) Refrigeration
 - B) Dehydration
 - C) Food Additives
 - D) Radiation
- 6.) One of the methods of preservation not mentioned above is *heat preservation*. Heat preservation uses heat (as the name states) to reduce the levels of microbes from growing. There are two categories of heat preservation methods: *mild* or *severe*. Which of the following statement(s) is (are) **incorrect** regarding the two methods? [JA]
- A) One disadvantage of severe heat treatment is the heavy alteration in textures and flavors.
 - B) Pasteurization is an example of severe heat treatment technique.
 - C) One advantage of mild heat treatment is that it requires another preservation method.
 - D) Products that underwent severe heat treatment have a higher shelf life compared to a product that underwent mild heat treatment.
 - E) The aim of severe heat treatment is to reduce bacterial count, but not to create a sterile product.
- 7.) The Food Scientists community always looks for a new way of preserving foods. One recently suggested method of preservation is Reduced oxygen packaging (ROP). Although it has advantages, it has one serious downfall. Which of the following is **the major concern regarding ROP**? [JA]
- A) It can alter the taste of the food.
 - B) It decreases the shelf life of the food.
 - C) It can facilitate the growth of *Clostridium botulinum*
 - D) It can increase fat oxidation.

Use the following diagram for Questions 8 to 10.

Amino acids can be organized into three categories: (1) Essential (2) Nonessential and (3) Conditionally essential. The table below shows the structure of 6 Amino Acids

		
A	B	C
		
D	E	F

8.) Which of the amino acids above are **essential**? (Select all that apply) [JA]

A) A

B) B

C) C

D) D

E) E

F) F

9.) Which of the amino acids above are **nonessential**? (Select all that apply) [JA]

A) A

B) B

C) C

D) D

E) E

F) F

10.) Which of amino acids above are **conditionally essential**? (Select all that apply) [JA]

A) A

B) B

C) C

D) D

E) E

F) F

Use the following information for Questions 11 to 15

There are many laboratory tests that can be conducted to determine the different types of compounds that exist within the food. Each test uses specific chemical reagents that will help the scientists to distinguish the types of nutrients. The three chemical tests are used: (1) **Iodine Test** (2) **Biuret Test** (3) **Benedict Test**.

11.) Which of the following compounds does the **Iodine Test** detect? [JA]

- A) Lipids
- B) Starch
- C) Proteins
- D) Monosaccharides
- E) Fatty Acids

12.) The complex interactions between what two compounds produce the blue-black color that indicates *positive* for starch? [JA] [TB #2]

- A) I_2 and amylose
- B) I_3^- and amylose
- C) I^- and amylose
- D) I_2 and amylopectin
- E) I_3^- and amylopectin
- F) I^- and amylopectin

13.) The steps below represent a simplified procedure of conducting a biuret test. Fill in the missing blank. [JA]

1. Add a few drops of copper sulfate solution to your food solution.
2. Add a few drops of sodium hydroxide solution
3. If the solution turns _____, the food contains protein.

- A) Black
- B) Blue
- C) Read
- D) Purple
- E) Orange

14.) Refer to the steps for the biuret test. Which of the following are **true** regarding the procedures and the results of the biuret test? (Select all that apply) [JA]

- A) Sodium Hydroxide is added to create an alkaline environment.
- B) The color change does not depend on the number of peptide bonds that are present in the food.
- C) The purpose of the sulfate ions is to add charge to the protein compounds so that Copper ions can bond with it.
- D) The color change from blue to pink represents the presence of amino acids.
- E) The color change can be contributed to the formation of complex coordination between Cu^{2+} and N atoms located in the amid bonds.

15.) Which of the following are components of Benedict's reagent? (Select all that apply) [JA]

- A) $Na_3C_6H_5O_7$
- B) $NaHCO_3$
- C) $Cu_3(PO_4)_2$
- D) $CuSO_4$
- E) $NaC_6H_5O_7$

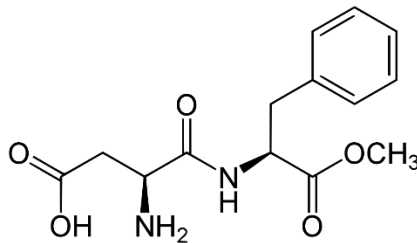
Use the following information for Questions 16 – 18

Artificial sweeteners are sugar substitutes with various degrees of sweetness, most of which are greater than table sugar in terms of magnitude. Although they produce the same *sweetness* as table sugars, they are distinct in terms of biochemical properties and characteristics.

- 16.) One of the biggest distinguishing characteristics between *sugar* and *artificial sweeteners* is their caloric value. Many artificial sweeteners are often noncaloric, meaning that they have zero calories. Which of the following artificial sweeteners are **noncaloric**? (Select all that apply) [JA]

A) Acesulfame K
B) Aspartame
C) Neotame
D) Saccharin
E) Sucralose

- 17.) The chemical structure below represents which artificial sweeteners? [JA]



A) Acesulfame K
B) Aspartame
C) Neotame
D) Saccharin
E) Sucralose

- 18.) Another category of sugar substitute is called *sugar alcohol*. Which of the following are characteristics of *sugar alcohol*? (Select all that apply) [JA]

A) They occur naturally in various fruits and vegetables.
B) They contain ethanol groups.
C) They are caloric.
D) They have a similar structure to glucose molecules.
E) They are not water soluble.

- 19.) What macronutrients are required to be on food labels? [LT]

A) Vitamin C, calcium, iron, and magnesium.
B) Vitamin D, sodium, iron, and potassium.
C) Vitamin A, sodium, iron, and magnesium.
D) Vitamin D, calcium, iron, and potassium.

- 20.) Let's get this bread! I have a 46 cm long baguette that weighs 454 grams, has a height of 4 cm, and has a density of 0.411 g/cm^3 . What is the breadth of my baguette? [LT]
- A) 75 cm^3
 - B) 0.17 cm^3
 - C) 6.0 cm^3
 - D) 14 cm^3
 - E) None of the above
- 21.) What temperature was this candy cooked to? Hint: think about what stage it probably had to be cooked to! (Refer to the image sheet) [LT]



- A) 230°F
 - B) 236°F
 - C) 300°F
 - D) 338°F
 - E) 350°F
- 22.) I am bored and want to order the water activity (a_w) of the following foodstuffs from low (1) to high (4). What is the correct order? (Refer to the image sheet for color) [LT]



- A.
 B.
 C.
 D.
- A) D B C A
 - B) B C D A
 - C) B D C A
 - D) A C D B
 - E) A D C B
 - F) C A D B

- 23.) What is tempering? [LT]
- A) The process of heating and cooling a chocolate mixture to rearrange the crystalline fat structure of the chocolate to increase the amount of stable beta crystals
 - B) The ability of sugar to delay staling in pastries and other food items due to its hygroscopic nature
 - C) The Hulk's somewhat unhealthy anger management method of binge eating because food fixes feelings
 - D) The process of breaking the fat within milk into tiny droplets that will then stay suspended within the fluid in an emulsion
 - E) None of the above

24.) Which sugar is the most hygroscopic? [LT]

- A) Glucose
- B) Fructose
- C) Galactose
- D) Sucrose
- E) Maltose
- F) Lactose

For Questions 25 to 27, use the following information.

Peaches were in season this summer, so last month my friend Eva decided to snack on some juicy, raw peaches! Almost immediately, she started feeling an itchy sensation in her mouth and noticed her lips were swelling slightly. She took an antihistamine and the symptoms disappeared within 30 minutes.

25.) What phenomenon was Eva probably experiencing? [LT]

- A) Cooties
- B) Anaphylactic shock
- C) Explosive diarrhea
- D) Oral allergy syndrome
- E) Myopia

26.) Which pollen are implicated in the oral cross-reactivity reactions with foods leading to the above phenomenon? [LT]

- A) Timothy and orchard grass.
- B) Mugwort
- C) Ragweed
- D) Birch
- E) All of the above
- F) None of the above

27.) People who suffer from this phenomenon after eating certain foods likely also suffer from seasonal allergies, also known as: (Select all that apply) [LT]

- A) Allergic rhinitis
- B) Leukemia
- C) Hay fever
- D) Yellow fever
- E) Hepatitis
- F) Irritable bowel syndrome

28.) Which of the following are included in the 8 most common food allergies in the USA? (Select all that apply) [LT]

- A) Pork
- B) Peanuts
- C) Tree Nuts
- D) Strawberries
- E) Milk
- F) Wheat
- G) None of the above

For Questions 29 to 31, use the following information.

I want to make dinner rolls for Thanksgiving, but am not sure how to achieve the light, fluffy, and soft texture of store-bought bread.

29.) What are the 3 leavening gases? (Select all that apply) [LT]

- A) Carbon monoxide
- B) Carbon dioxide
- C) Helium
- D) Air
- E) Steam
- F) None of the above

30.) I decided to try a recipe for yeast rolls and noticed that my rolls were slightly flattened and didn't have the nice, domed tops that they should. What could have gone wrong? [LT]

- A) Not enough water added to the dough
- B) Insufficient proofing
- C) Too much air beat into the dough
- D) Too much butter made the gluten too slippery
- E) Not enough milk added to the dough

31.) Hmm, since the yeast rolls were a bust, maybe I should try leavening with baking soda! Which of the following ingredients would be required in a batter in order for baking soda to act? [LT]

- A) Yeast
- B) Sugar
- C) Eggs
- D) Milk
- E) Brown sugar
- F) None of the above

Use the following information for Questions 32 to 33

Nutrition Facts	
80 servings per container	
Serving size 2 oz (56g)	
Amount per serving	
Calories 160	
% Daily Value*	
Total Fat 11g	14%
Saturated Fat 4g	20%
Cholesterol 50mg	17%
Sodium 290mg	13%
Total Carbohydrate 2g	1%
Total Sugars 2g	
Includes 2g Added Sugars	4%
Protein 13g	
Iron 0.4mg	2%
Potassium 90mg	2%
Not a significant source of trans fat, dietary fiber, vitamin D, and calcium.	
* The % Daily Value (DV) tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.	
INGREDIENTS: Smoked Cooked Sugar, Salt, Spices.	

- 32.) What 2 vitamins are no longer required to be on a nutrition label, according to the new regulations (in other words, they were once mandatory but are now optional to include on nutrition labels)? [LT]
- A) Vitamin A
 - B) Vitamin B₁₂
 - C) Vitamin C
 - D) Vitamin D
 - E) Vitamin E
 - F) N/A; the required vitamins have not changed.
- 33.) What 2 items on the new nutrition food label are now required to be in a large, bolded font? (Select all that apply) [LT]
- A) Nutrition Facts
 - B) Sodium
 - C) Serving size
 - D) Calories
 - E) Cholesterol
 - F) N/A; it is up to the manufacturer to decide what is bolded.

Use the following information for Questions 34 to 36

Starches are used in cooking to thicken liquids, such as in soups, stews, puddings, and pastry fillings.

- 34.) The increase of volume, viscosity and translucency when starch is heated with a liquid is called:
- A) Syneresis
 - B) Gelatinization
 - C) Retrogradation
 - D) Dextrinization
 - E) None of the above
- 35.) During the process discussed in the previous question, water and heat break the bonds within starch molecules, allowing water to move in and form hydrogen bonds within the starch molecule. This causes the starch granules to swell and soften, and if the starch has a high enough amylose content it may even form a semi-solid paste in a process called:
- A) Catabolism
 - B) Rancidity
 - C) Gelation
 - D) Proofing
 - E) Emulsion
 - F) None of the above
- 36.) Which of the following are starches commonly used as thickening agents in cooking? Select all that apply.
- A) Arrowroot
 - B) Buckwheat
 - C) Cornstarch
 - D) Sweet rice flour
 - E) None of the above

Section II: 4 Free Response Questions

1- The Citric Acid Cycle (15 points) [JA]

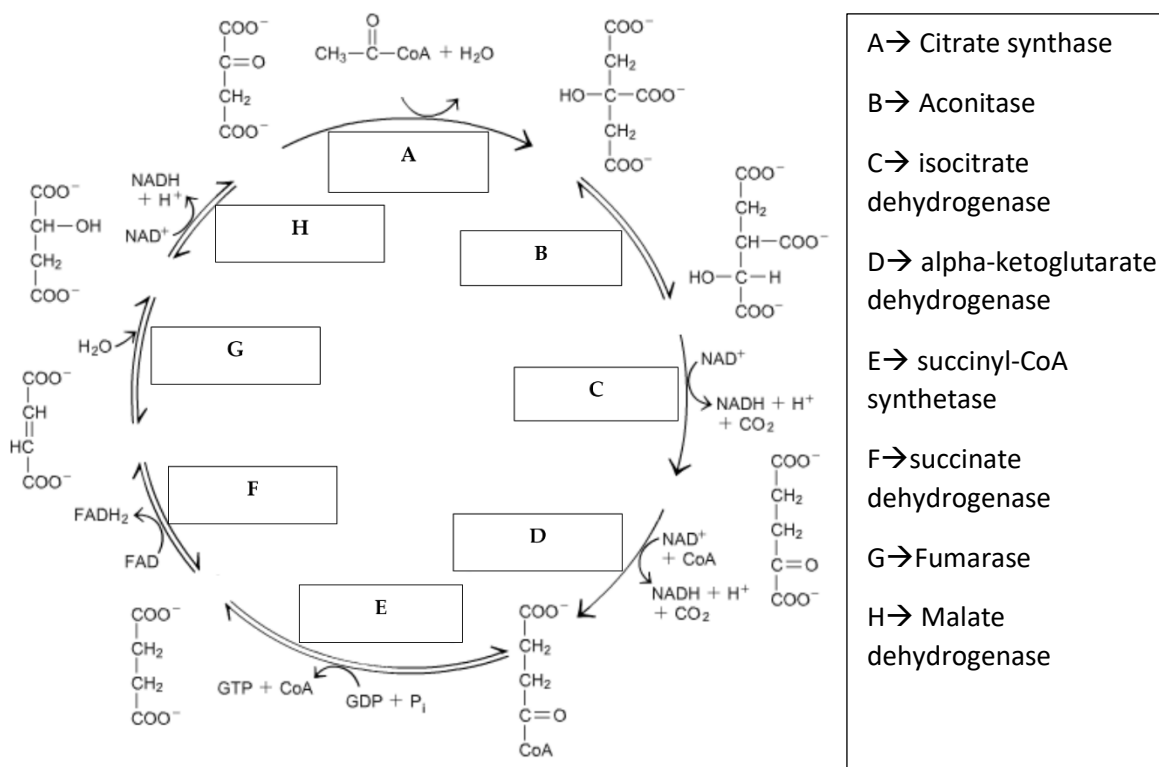
According to the best biology book *Campbell Biology* (in my opinion :D), the citric acid cycle is described as a *metabolic furnace that further oxidizes organic fuel derived from pyruvate*. As the book states, the citric acid cycle plays an important role in reorganizing the energy back to certain molecules that can be used in oxidative phosphorylation. Without the citric acid cycle, the human body will not be able to produce important energy carrying molecules.

(a) Let's start with the purpose of the citric acid cycle. What exactly is the purpose of this long cycle? (1 point)

Example Answer: To produce high energy compound such as ATP and energy rich reduced coenzymes (NADH & FADH₂)

- Must mention ATP, NADH, and FADH₂ for 0.50 points
- Another 0.50 points for stating they are *high energy compounds*.

(b) Label the **enzymes** shown in the diagram below. Each label is worth 0.50 points. (4 points) [TB #3]



(c) How many NADH, FADH₂, and ATP are produced when 4 glucose molecules are used for glycolysis? (3 points)

- 1 point per compound (Work does NOT need to be shown)
 - o 24 NADH, 8 FADH₂, and 8 ATPs

(d) Throughout the cycle, two carbon atoms are lost. As what compounds do they exist the cycle and are they the same two carbons that entered as acetyl-CoA? (2 points)

- 1 point for leaving as CO₂
- 1 point for NO, they are not the same

(e) Which of the compounds in the citric acid cycle are **oxidized** and which are **reduced**? (2 points)

- Isocitrate, Malate and Alpha ketoglutarate are oxidized. (1.5 points)

- NAD^+ is reduced to $NADH$ and FAD is reduced to $FADH_2$ (0.5 points – All or nothing)

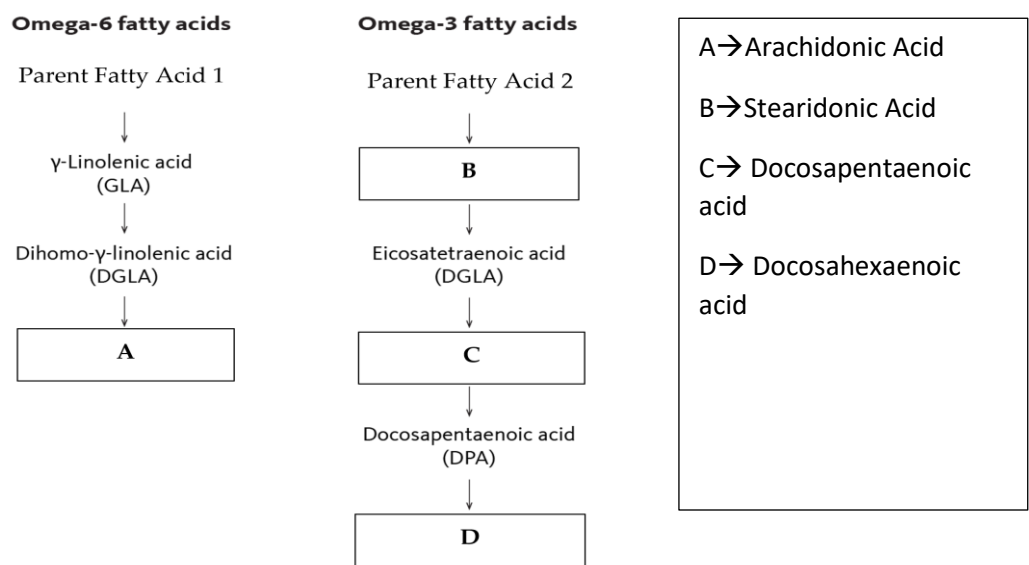
(f) The ATP generated during the Citric Acid Cycle is done through the mechanism called **substrate-level phosphorylation**. Describe this process. (1 point)

- Substrate-level phosphorylation is a reaction that forms ATP by transferring a phosphate group from a substrate directly to ADP or GDP.
- 0.50 point for mentioning formation of ATP
- 0.50 for mentioning *direct transfer of a phosphate group*.

2-) Essential Fatty Acids (10 points) [JA]

Fatty acids are the building blocks of the macromolecule *lipid*. It consists of a long chain of nonpolar tails and a carboxylic acid. Without fatty acids, our body will not be able to synthesize membranes and other important structures. A group of *fatty acids* is called *essential* as they cannot be produced naturally by the human body. Thus, we must obtain them from other food products in order to sustain the recommended value.

- (a) What **two** fatty acids are considered to be **essential**? (2 points)
- Alpha Linoleic Acid (1 point)
 - Linolenic Acid (1 point)
- (b) All humans and mammals have the elements and materials that are necessary to build the essential fatty acids compounds. However, they lack certain key components that configure the elements into the essential fatty acids' structures. State the key components and why they are important. (4 points) [TB #4]
- Humans lack specific **enzymes** (specifically delta 12 and delta 15 desaturase) that can insert a **cis** double bond at the *n*-3 and *n*-6 positions of the fatty acids. (which is necessary to produce the omega-6 and omega-3 structures)
 - (+1) for mentioning the lack of enzymes
 - (+1) for mentioning the specific names.
 - (+2) for describing its role in adding a **cis** double bond at the two sites (*n*-3 and *n*-6)
- (c) The diagram below shows the longer chain derivatives of the two parent fatty acids. Fill in the missing spaces. (2 points)



- (d) Some fatty acids are *conditionally essential*. What does the term *conditionally essential* mean? (1 point)

Conditionally essential means that certain fatty acids can become essential under certain disease conditions. This could also be the case when certain conditions may result in the demand exceeding the body's capacity for synthesis

- (+1) for why it is *conditionally essential* – certain conditions causing excessive amount of demand.
- (e) There are various clinical signs of essential fatty acid deficiency. Some of them are listed below. Choose the correct word to fill in the blank. (1 point)
- (i) _____ rash (moisture | **dry**)
- (ii) _____ growth in infants (increased | **decreased**)

3-) Willy Wonka [LT]

My boy Willy Wonka is the best candymaker in the multiverse, so when I was procrastinating way too long on writing this exam I asked him to help me write some questions for y'all!

The concentration of sugar within a cooked sugar solution is inversely proportional to the temperature of the solution. (1 pt)

True / False

Candies can be classified into 2 main categories, crystalline candies and non-crystalline candies. What are 2 ways crystallization is prevented in non-crystalline candies? (4 pts)

In non-crystalline candies, crystallization is prevented ...

1) By adding a large amount of interfering agents (1 pt) to limit agitation, thus preventing the formation of nuclei for crystallization (1 pt)

2) By concentrating the sugar solution (1pt) so that the mixture will cook quickly (1 pt) due to its high viscosity.

23. Which of the following is a non-crystalline candy? Select all that apply.

A) Fudge

B) Lemon drops

C) Fondant

D) Cotton candy

E) Gummies

F) None of the above

4-) Egg [LT]

Rocky Balboa eats 8-11 raw eggs every single day and develops a biotin (vitamin B₇) deficiency. This is because there is a protein called avidin in raw egg whites that binds to biotin, making it biologically unavailable for the body to absorb. Why do people who eat cooked eggs not develop a biotin deficiency?

Cooking the eggs **denatures the protein** avidin, which frees the biotin and makes it available for absorption

Section III: True or False

- 1-) Oxidoreductase is a class of enzyme [JA] (TRUE)
- 2-) Maltose is a disaccharide formed by the hydrolysis of starches [JA] (TRUE)
- 3-) Cellulose's monomers are linked by α 1 – 4 glycosidic linkages [JA] (FALSE)
- 4-) The solidification of the egg white when boiled has to do with protein denaturation [JA] (TRUE)
- 5-) Ribozyme is composed of proteins. [JA] (FALSE)
- 6-) The FDA increased the recommended daily calorie intake to 2,200 calories per day. [LT] (FALSE)
- 7-) Baking soda requires a liquid and an acid in order to produce carbon dioxide. [LT] (TRUE)
- 8-) When cooking a sugar solution to create crystalline candy, adding an *invert sugar* can discourage the premature formation of crystals [LT] (TRUE)
- 9-) Beef jerky has a higher water activity than raw carrots [LT] (FALSE)

Section IV: Hands-on Lab Practical

Directions: There are 2 beakers filled with sugar water at each lab station around this room. Go to one station and determine the sugar content in each sample using your hydrometer. Write down the percent concentration by mass of the sugar in the solutions.

1.) Sugar content of Beaker 1: 3%

2.) Sugar content of Beaker 2: 6%